

Partition Draws, Not Aggregators: Decomposing Result Variance in Federated LoRA Fine-Tuning Under Non-IID Splits

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Abstract—Federated LoRA comparisons often rest on a single non-IID partition draw, even when reported method gaps sit near ordinary run-to-run noise. We ask how much of the variation in held-out next-token loss comes from the partition draw, how often few-draw rankings flip, and how many draws a target effect size requires. In a pre-registered design of 120 runs at two scales (TinyLlama-1.1B-Chat and LLaMA-3.2-3B), with partition seeds separated from training seeds, we fit a two-way variance decomposition for FedIT, FFA-LoRA, and FLoRA on Dolly, partitioned by its category labels at Dirichlet $\alpha=0.1$. On TinyLlama the partition and interaction shares are 0.508 and 0.469 against a residual of 0.023. On LLaMA-3.2-3B the interaction dominates (0.764) and the partition interval includes zero. Ranking flips concentrate in the near-tied FedIT-FLoRA pair: the three-draw paired flip probability is 0.377, while gaps of about 0.023 never reverse in a single draw. Detecting that observed near-tie at 80% power needs 206 paired partition draws on TinyLlama and 46 on LLaMA-3.2-3B; separable pairs need only 3. We recommend reporting the number of partition draws, pairing comparisons on partitions, and stating the smallest difference the design can detect. We make no method recommendation.

Keywords—federated learning; low-rank adaptation; non-IID partitioning; experimental variance; reproducibility; large language models

I. INTRODUCTION

Federated LoRA methods for instruction tuning report method differences that are often small relative to the noise in a single evaluation. FFA-LoRA [1] publishes means over 20 runs with standard deviations; in the non-private setting the LoRA standard deviation on MNLI-matched (10.7) exceeds the mean gap to FFA-LoRA (3.02 points). FLoRA [2] compares to FedIT [3] with single reported MMLU and MT-bench values after 1 to 3 communication rounds, without reported seed replication, including a Llama/Alpaca MMLU margin of 0.44 points that the paper itself calls “marginal.” Those practices establish the motivating fact: published federated LoRA contrasts can sit inside, or only slightly above, ordinary run-to-run variation. What they do not establish is how much of that variation comes from the non-IID partition draw itself.

In Dirichlet or category-based federated simulations, the “non-IID” condition is commonly instantiated from a single partition seed: the Dirichlet construction itself [4] and the partition catalogues built on it [5] treat the draw as a setting rather than a source of variance, and the federated LoRA evaluations above report no resampling of it. In the evaluations cited above, training seeds are sometimes repeated while the partition seed is not. If held-out loss moves with the partition

draw by as much as it moves with the training seed, then a ranking obtained from one or two partitions is not a property of the methods alone. The open question is quantitative: how large is the partition component, how often do few-draw rankings flip, and how many draws are needed for a target effect size?

We answer three pre-registered questions on FedIT, FFA-LoRA, and FLoRA, using held-out next-token loss on Dolly as the response. All differences are in held-out next-token loss, where the unadapted base models score 2.121 and 2.169 and fine-tuned runs fall between 1.69 and 1.84. The design freezes the federation (ten clients, full participation, fifteen rounds, one local epoch), the LoRA configuration, and the analysis plan before any production run, and separates partition seeds from training seeds at two model scales.

- **RQ1.** At Dirichlet $\alpha=0.1$, what share of the variance comes from the partition draw, the partition-by-method interaction, and the training seed?
- **RQ2.** How often does an evaluation using k partition draws rank the three methods differently from the full-design ranking?
- **RQ3.** How many partition draws are needed to detect held-out loss differences of 0.005, 0.01, 0.02, and 0.05 at 80% power, for paired and unpaired designs?

An exploratory RQ4 on partition statistics is reported but not used for claims.

a) Contributions.: (1) A pre-registered 120-run design at two scales (TinyLlama-1.1B-Chat and LLaMA-3.2-3B) that separates partition seeds from training seeds under a frozen federation and LoRA configuration. (2) A method-of-moments variance decomposition with cluster-bootstrap intervals, showing on TinyLlama that partition and interaction shares dominate the residual (0.508 and 0.469 versus 0.023), while on LLaMA-3.2-3B the interaction dominates (0.764) and the partition main-effect interval includes zero. (3) A ranking-stability analysis conditioned on effect size: the near-tied FedIT-FLoRA pair flips under few draws (three-draw paired flip probability 0.377 on TinyLlama), whereas gaps of about 0.023 to 0.031 never reverse in a single draw. (4) A draws-needed table for pre-registered deltas and for the observed gaps, including 206 and 46 paired draws for the near-tie at the two scales, and the observation that where the interaction dominates, pairing on partitions buys little. We recommend reporting the number of partition draws, pairing comparisons on partitions, and stating the smallest difference the design can detect. We make no method recommendation.

Section II situates the work against benchmark-variance studies, non-IID simulation practice, and federated LoRA evaluation. Section III states the design and variance model. Section IV reports the measurements. Section V gives reporting recommendations and limitations.

II. RELATED WORK

a) Variance and seed sensitivity in ML benchmarks.:

Empirical comparisons are sensitive to sources of variation beyond the method under test. Bouthillier et al. [6] model data sampling, initialization, and hyperparameter choice as variance components in machine-learning benchmarks and show that ignoring them inflates the chance of declaring a false improvement; their recommendations include randomizing multiple sources of variation and accounting for the resulting uncertainty when claiming gains. Picard [7] isolates the random seed alone: across 10,000 CIFAR-10 trainings the accuracy range ran from 89.01% to 90.83% (1.82 percentage points), with about 0.5% on ImageNet. Both lines of work motivate multi-seed summaries and caution around small margins. Neither separates the draw of a federated data partition from the training seed, which is the decomposition this paper targets.

b) Non-IID simulation in federated learning.:

Federated evaluations routinely induce client heterogeneity by construction. Hsu et al. [4] synthesize label skew by drawing each client's class proportions from a Dirichlet distribution $\text{Dir}(\alpha p)$, with smaller α producing more skewed clients, and show that FedAvg accuracy degrades as α decreases. Natural partitions that follow writers, users, or categories appear in benchmarks such as LEAF [8]. NIID-Bench [5] compares algorithms under six partitioning strategies and reports that no existing algorithm outperforms the others in all cases. Federated instruction-tuning work inherits these simulation habits: FedIT [3] splits Databricks-dolly-15k into ten shards, one per client, and reports per-client category distributions in which several categories are absent entirely, and FFA-LoRA [1] uses a 3-client cross-silo setting with fixed label proportions (for example $[0.1, 0.9]$ and $[0.9, 0.05, 0.05]$), not a Dirichlet draw, on RoBERTa/GLUE and LLaMA/GSM-8K rather than Dolly. None of the evaluations above resamples the partition seed; the single draw is treated as the non-IID condition, even though that draw determines which clients are empty of a category or too small to take an optimizer step.

c) Federated LoRA methods and evaluation practice.:

FedIT [3] aggregates LoRA adapters with FedAvg for instruction tuning. FFA-LoRA [1] freezes A and trains B , reporting accuracies over 20 runs as mean and standard deviation. In the non-private setting the MNLI-matched means are 85.05 ± 1.1 (FFA-LoRA) versus 82.03 ± 10.7 (LoRA), and the QQP means are 84.35 ± 0.6 versus 83.51 ± 3.3 ; the LoRA standard deviation of 10.7 on MNLI-matched exceeds the mean method gap, which is published evidence that run-to-run variance in federated LoRA can dominate the reported contrast. FLoRA [2] stacks heterogeneous-rank adapters and compares to FedIT on MMLU and MT-bench with 10 clients under FedIT's non-IID setting (matching our client count). Several margins are modest: Llama/Alpaca MMLU is 29.85 versus 29.41, which the paper calls "marginal," and homogeneous MT-bench gains are described as at least 0.2. Those runs use only 1 to 3 communication rounds with single reported values and no seed

replication, and MMLU is scored on a 1,444-sample subset. Published evaluations therefore sometimes report across-run means and standard deviations, yet they do not resample the partition draw or decompose that run variance into partition and training-seed components when the reported margin is on the order of a point or less.

d) *Gap.*: To our knowledge, no prior study decomposes partition-draw variance against training-seed variance for federated LoRA instruction tuning, or tests whether method rankings survive that partition variance. This paper fills that gap under a pre-registered design.

III. EXPERIMENTAL SETUP

We study three federated LoRA aggregation methods that appear in recent instruction-tuning work: FedIT [3] (FedAvg over LoRA adapters), FFA-LoRA [1], and FLoRA [2]. No other aggregators are included. The design is fixed before production runs and is identical across methods except for the aggregation rule itself.

a) *Models and adapters.*: The primary model is TinyLlama-1.1B-Chat-v1.0. The replication model is LLaMA-3.2-3B, retained after a pre-registered timing gate on an RTX 4090 measured 65.2 minutes for a full run with a peak allocation of 9.5 GB, within the gate's limits of 75 minutes and 22 GB. Both models use the same LoRA configuration: rank $r=16$, scaling $\alpha_{\text{LoRA}}=32$, dropout 0.1, and target modules restricted to `q_proj` and `v_proj`. The base model is frozen; only the LoRA parameters are trained and communicated. Both models use learning rate 1×10^{-4} , weight decay 0.01, and maximum sequence length 256.

b) *Federation.*: Each run uses 10 clients with full participation for 15 rounds and one local epoch per round. TinyLlama uses batch size 4 with gradient accumulation 4; LLaMA-3.2-3B uses batch size 2 with gradient accumulation 8. The client training loop steps the optimizer only on complete gradient-accumulation blocks and does not use `drop_last`, so clients with fewer than 13 samples (TinyLlama) or 15 samples (LLaMA) contribute zero optimizer steps. In the Dirichlet $\alpha=0.1$ partitions, effective clients range from 7 to 10 on TinyLlama and from 9 to 10 on LLaMA-3.2-3B; discarded trailing samples per partition seed range from 27 to 60 (TinyLlama) and 41 to 104 (LLaMA); client sizes range from 2 to 1218; and active clients take only the values 9 or 10.

c) *Data and partitions.*: All training examples come from Databricks Dolly-15k, using `train[0:3000]`. The non-IID setting is Dirichlet label skew over the dataset's real `category` column at $\alpha=0.1$. The IID setting uses a single fixed data seed. Partition draws are controlled by data seeds that are separate from the training (run) seeds. TinyLlama uses data seeds 2001–2010 with run seeds $\{7001, 7002\}$ at $\alpha=0.1$ (60 cells) and data seed 2001 with run seeds 7001–7005 for IID (15 cells). LLaMA-3.2-3B uses data seeds 2001–2006 with the same two run seeds at $\alpha=0.1$ (36 cells) and data seed 2001 with run seeds 7001–7003 for IID (9 cells). The total design is 120 production cells.

d) *Held-out metric.*: The primary metric is held-out next-token loss on Dolly `train[3000:3500]` (500 examples),

maximum sequence length 256, using the same prompt format as training, including a Context field when present (formatter version v2-dolly-context). Base (unadapted) held-out loss is 2.120666 on TinyLlama and 2.168946 on LLaMA-3.2-3B and is identical across all runs of a given model. TinyLlama holdout evaluation runs in float32. LLaMA-3.2-3B production holdouts run in float16 with sequential loading (the tuned adapter is freed before the base model is loaded) because two concurrent float32 3B copies exceed 24 GB. A float32 spot-check on three LLaMA cells spanning the production tuned-loss range found a maximum absolute difference of 4.1×10^{-5} relative to float16, about 250 times below the 0.01 effect scale used for power calculations, so float16 was retained for the LLaMA production numbers.

e) Communication.: Secondary reporting uses measured communication in megabytes, counting the tensors actually transmitted in each direction following the byte accounting in [9]. Per-client upload payloads match the frozen constants: 9,011,200 bytes for TinyLlama (FedIT and FLoRA) and 9,175,040 bytes for LLaMA-3.2-3B (FedIT and FLoRA). FFA-LoRA, as implemented in this codebase, uploads both *A* and *B* on the first round and *B* thereafter, and downloads *B* only (3,244,032 bytes per client on TinyLlama). Total communication therefore differs across methods by construction. We report communication only within method across partitions and do not rank methods by bytes.

f) Hardware and software provenance.: Every production cell ran on a single NVIDIA GeForce RTX 4090 with one worker process per GPU. Median wall-clock time is 27.1 minutes for TinyLlama and 51.1 minutes for LLaMA-3.2-3B. Training provenance is the `freeze-v1` tag with a clean tracked tree. The LLaMA float16 holdout change is an eval-only amendment tagged `freeze-v2`; it does not alter training code or aggregated adapters. The design, configurations, grids, and analysis plan were committed and tagged before any production run, and the analysis follows that plan; deviations are logged in the repository's decision log and reported here. Three cells resumed from a checkpoint after host hangs (one TinyLlama FLoRA cell and two LLaMA FLoRA cells); their held-out losses fall inside the range of the other run seeds that share the same grid, heterogeneity, data seed, and method.

g) Variance model.: For each model we fit a two-way design with methods fixed ($m=3$), partitions random ($p=10$ for TinyLlama, $p=6$ for LLaMA-3.2-3B), and $r=2$ training seeds per cell. Method-of-moments estimates from the mean squares give the residual (training-seed) component $\hat{\sigma}_E^2$, the partition-by-method interaction $\hat{\sigma}_{PM}^2$, and the partition component $\hat{\sigma}_P^2$, with negative estimates truncated at zero and truncation reported. Confidence intervals come from a cluster bootstrap that resamples partitions with replacement ($B=2000$). A mixed-model fit serves as a cross-check; the moment estimates are primary. Ranking stability uses $B=10,000$ resamples of k partition draws under paired and unpaired protocols, and required sample sizes come from a two-sided t test at $\alpha=0.05$ with 80% power. All random procedures use a fixed seed.

IV. RESULTS

All analyses use held-out loss as the response. Numbers below come from the pre-registered tables in `analysis/`.

TABLE I. FROZEN EXPERIMENTAL DESIGN.

Element	Value
Methods	FedIT, FFA-LoRA, FLoRA
Models	TinyLlama-1.1B-Chat-v1.0; LLaMA-3.2-3B
LoRA	$r=16$, $\alpha_{\text{LoRA}}=32$, dropout 0.1, $q_{\text{proj}}+v_{\text{proj}}$
Federation	10 clients, full participation, 15 rounds, 1 local epoch
Data	Dolly-15k train[0:3000]; held-out train[3000:3500]
Non-IID	Dirichlet label skew on category, $\alpha=0.1$
IID	Fixed data seed split
Primary metric	Held-out loss (formatter v2-dolly-context)
Secondary	Measured communication (MB)
TinyLlama cells	60 ($\alpha=0.1$) + 15 (IID)
LLaMA-3.2-3B cells	36 ($\alpha=0.1$) + 9 (IID)

We report TinyLlama and LLaMA-3.2-3B separately and do not recommend a method.

a) Variance components at $\alpha=0.1$.: Table II summarizes the method-of-moments estimates under the two-way design in Section III. On TinyLlama ($p=10$), the partition share is $\hat{\sigma}_P^2/(\hat{\sigma}_P^2+\hat{\sigma}_{PM}^2+\hat{\sigma}_E^2)=0.508$ with bootstrap 95% CI [0.202, 0.663], the interaction share is 0.469 [0.316, 0.765], and the residual share is 0.023 [0.007, 0.056]. Absolute components are $\hat{\sigma}_P^2=4.09 \times 10^{-6}$, $\hat{\sigma}_{PM}^2=3.78 \times 10^{-6}$, and $\hat{\sigma}_E^2=1.89 \times 10^{-7}$. Partition and interaction are comparable in magnitude; residual training-seed noise is an order of magnitude smaller. No estimate was truncated at zero. The pre-registered mixed-model cross-check recovers the residual component on both models (1.89×10^{-7} and 1.31×10^{-7} , matching $\hat{\sigma}_E^2$) and a method-level variance component comparable to $\hat{\sigma}_P^2+\hat{\sigma}_{PM}^2$, but the separate partition component did not identify under the specified variance-component formula. We therefore report the moment estimates as primary, as pre-registered, and treat the mixed model as a partial cross-check.

On LLaMA-3.2-3B ($p=6$), the interaction dominates: share 0.764 [0.443, 0.990], with $\hat{\sigma}_{PM}^2=1.11 \times 10^{-5}$. The partition share is 0.227 [0.000, 0.551] and the residual share is 0.009 [0.002, 0.043], with $\hat{\sigma}_P^2=3.28 \times 10^{-6}$ and $\hat{\sigma}_E^2=1.31 \times 10^{-7}$. Because the partition CI includes zero, we do not claim a partition main effect at 3B; the 3B story is the interaction. Cross-scale bootstrap CIs on the partition share overlap ([0.202, 0.663] versus [0.000, 0.551]), so the data do not distinguish the two scales on that margin. No estimate was truncated at zero.

b) IID noise floor.: Pooling the per-method IID variances gives 6.40×10^{-8} on TinyLlama and 3.07×10^{-7} on LLaMA-3.2-3B. The ratio of $\hat{\sigma}_E^2$ at $\alpha=0.1$ to this IID floor is 2.96 on TinyLlama. On LLaMA the ratio is 0.43, but each method's IID variance has only two degrees of freedom (three run seeds), and the per-method IID variances differ by more than an order of magnitude (2.4×10^{-8} for FFA-LoRA versus 6.6×10^{-7} for FLoRA). We therefore report the 3B IID floor descriptively and do not interpret the ratio below one as evidence that $\alpha=0.1$ is quieter than IID.

c) Ranking stability.: The reference ranking at $\alpha=0.1$ orders methods by mean held-out loss over all cells: TinyLlama ranks FLoRA, FedIT, then FFA-LoRA; LLaMA ranks FedIT, FLoRA, then FFA-LoRA. Under the pre-registered paired protocol ($B=10,000$), the probability that the best method differs from this reference equals the probability that the full order differs (the two coincide because the only pair whose order

changes is the top one; the third method is separated by an order of magnitude more than the per-draw noise). On TinyLlama those probabilities are 0.555, 0.406, 0.377, and 0.284 for $k \in \{1, 2, 3, 5\}$ (Wilson 95% CI half-widths about 0.01). On LLaMA they are 0.168, 0.272, and 0.287 for $k \in \{1, 2, 3\}$. Unpaired probabilities are similar in magnitude and are shown in Figure 2 and `analysis/rank_flip.csv`; the paired protocol is primary.

The aggregate flip rate is entirely the near-tied FedIT-FLoRA pair. On TinyLlama that pair has true mean gap $+0.00055$; single-draw sign flips occur in 11 of 20 cells, and the paired order-flip probability is 0.555 at $k=1$ and 0.377 at $k=3$. The separable pairs (FedIT versus FFA-LoRA at -0.023 , FFA-LoRA versus FLoRA at $+0.024$) never reverse: 0 of 20 single draws and paired flip probability 0 at every k . On LLaMA, FedIT versus FLoRA has gap -0.0020 with 2 of 12 single-draw sign flips and paired flip probabilities 0.168 ($k=1$) and 0.287 ($k=3$); the separable gaps -0.033 (FedIT versus FFA-LoRA) and $+0.031$ (FFA-LoRA versus FLoRA) never flip (0 of 12). The rise in LLaMA flip probability from $k=1$ to $k=3$ is a small-sample artifact of six partitions and a near-tie, not evidence that more draws are worse.

Conditioned on effect size, the ranking claim is therefore: pairs separated by less than the per-draw paired SD (0.00282 on TinyLlama, 0.00473 on LLaMA; empirical means 0.00276 and 0.00461) are ordered near-randomly by a few-partition evaluation, while gaps of about 0.023 (TinyLlama) and 0.031 (LLaMA) were recovered in every single draw. The flip table is a restatement of that conditioning, not a separate claim about method quality.

d) Draws needed.: Using the model-based paired SD, detecting a pre-registered $\Delta=0.005$ at 80% power (two-sided $\alpha=0.05$) requires 5 paired partition draws on TinyLlama and 10 on LLaMA; $\Delta=0.01$ requires 3 and 5; $\Delta=0.02$ requires 3 and 3; $\Delta=0.05$ requires 2 and 3. Unpaired per-method n values are similar (TinyLlama 7, 3, 2, 2; LLaMA 11, 4, 3, 2). For the observed gaps, the near-tie would need 206 paired draws on TinyLlama and 46 on LLaMA, whereas each separable pair needs only 3 paired draws on either model. That contrast is the operational form of the ranking result: a few partitions suffice when the gap is large relative to the paired SD, and hundreds are required when it is not.

e) Observed method differences.: Table III reports means and SDs. At $\alpha=0.1$, TinyLlama means are FLoRA 1.6957 (SD 0.0013), FedIT 1.6962 (0.0038), and FFA-LoRA 1.7192 (0.0026), each over $n=20$ cells; LLaMA means are FedIT 1.8028 (0.0051), FLoRA 1.8049 (0.0020), and FFA-LoRA 1.8355 (0.0031), each over $n=12$. Paired t -tests on partition-level means with Holm correction reject FFA-LoRA versus each of the other two methods on both models (TinyLlama Holm $p < 10^{-8}$; LLaMA Holm $p < 10^{-4}$) and do not reject FedIT versus FLoRA (TinyLlama Holm $p = 0.57$; LLaMA Holm $p = 0.41$). The FedIT-FLoRA confidence intervals include zero on both models. We report these contrasts as context for the ranking and power sections and do not claim a winning method.

f) Partition statistics and communication.: Among five exploratory Spearman correlations of partition-mean held-out loss with partition statistics, TinyLlama mean Jensen-Shannon

divergence has $\rho=0.72$ ($p=0.019$, $n=10$). That association does not survive Holm correction across the five TinyLlama tests (Holm $p=0.093$) and does not replicate on LLaMA ($\rho=0.086$, $p=0.87$, $n=6$). Active clients vary only from 9 to 10 and show no reliable association ($\rho=0.35$, $p=0.32$ on TinyLlama; $\rho=0.13$, $p=0.80$ on LLaMA). We treat RQ4 as exploratory and do not build an argument on any single correlation.

Measured total communication matches the V6 payload formula exactly for all six method-model pairs ($\max|\text{error}|=0$). Across $\alpha=0.1$ partitions, TinyLlama totals range from 2320.3 to 2578.1 MB for FedIT and FLoRA and from 884.8 to 983.1 MB for FFA-LoRA; LLaMA ranges are 2362.5 to 2625.0 MB and 992.25 to 1102.5 MB. The FFA-LoRA range is lower because this implementation transmits A and B only in the first round and B alone thereafter, in both directions; we report the values as measured within method and do not rank methods by bytes.

g) Pre-registered claim mapping.: Following the outcome-to-claim table, TinyLlama triggers row 1 on the partition share (0.508). LLaMA triggers row 1 only through the three-draw flip probability (0.287), with the near-tie caveat above, and additionally triggers row 4 because the interaction share (0.764) exceeds the partition share. Cross-scale partition-share CIs overlap (row 6). No other pre-registered claim rows are selected for either model.

V. DISCUSSION

a) What to report.: A multi-seed mean alone does not answer whether a federated LoRA ranking is stable under the non-IID draw. FFA-LoRA already shows that across-run standard deviations can exceed method gaps [1]; our results go one step further by decomposing that variation into a partition component, a partition-by-method interaction, and a residual training-seed component. At Dirichlet $\alpha=0.1$, the residual $\hat{\sigma}_E^2$ is small relative to $\hat{\sigma}_P^2$ and $\hat{\sigma}_{PM}^2$ on TinyLlama, so reporting only run seeds understates the uncertainty that matters for ranking. Practical reporting should therefore (i) separate the partition seed from the training seed, (ii) publish the variance shares or at least the paired per-draw SD of method differences, and (iii) state how many partition draws were used. When the observed gap is near that paired SD (0.00282 on TinyLlama, 0.00473 on LLaMA-3.2-3B), a few-draw evaluation is nearly a coin flip; when the gap is about 0.023 to 0.031, three paired draws sufficed here.

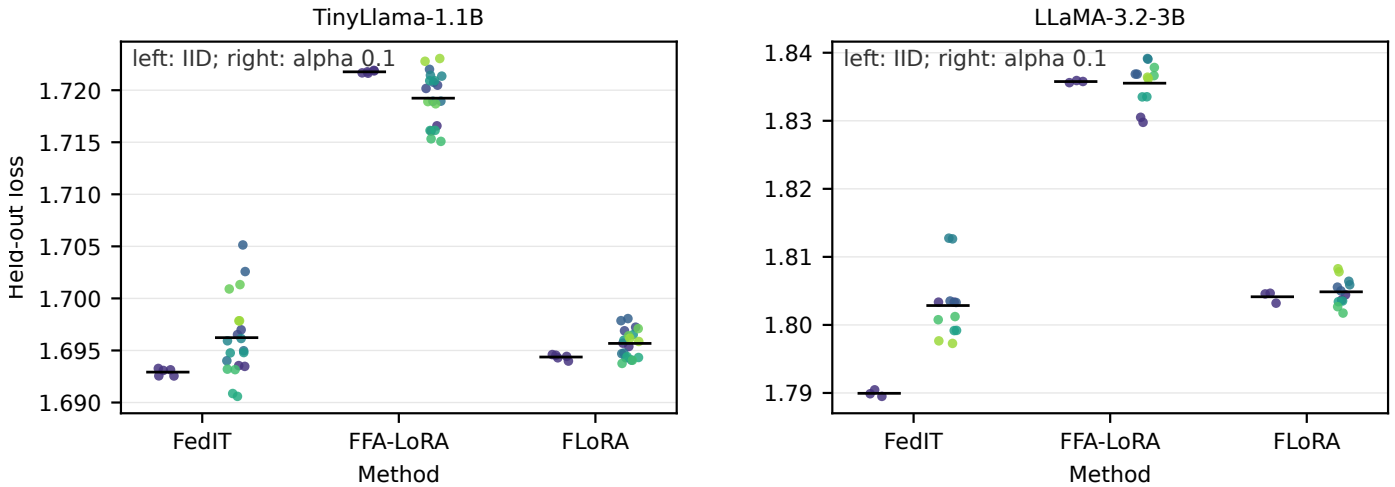
b) What pairing fixes, and what it does not.: Pairing methods on the same partition draws removes the shared partition main effect from the comparison and is the right default for ranking. It does not remove the interaction: methods can still reorder across draws when their responses to a partition differ. That is exactly the FedIT-FLoRA near-tie ($+0.00055$ on TinyLlama, -0.0020 on LLaMA), where best-method and full-order flip probabilities coincide because only the top pair moves. Pairing also does not invent power. Detecting a pre-registered $\Delta=0.005$ still needs 5 (TinyLlama) or 10 (LLaMA-3.2-3B) paired draws, and the observed near-tie would need 206 and 46. For that near-tie, pairing roughly halves the requirement at 1.1B (206 versus 416 per method unpaired) but helps only modestly at 3B (46 versus 57). At some larger

TABLE II. METHOD-OF-MOMENTS VARIANCE COMPONENTS AT DIRICHLET $\alpha=0.1$ WITH BOOTSTRAP 95% CIs OVER PARTITIONS ($B=2000$).

Model	$\hat{\sigma}_P^2$	$\hat{\sigma}_{PM}^2$	$\hat{\sigma}_E^2$	Share P	Share PM	Share E	p	m	r
TinyLlama-1.1B	4.09e-06	3.78e-06	1.89e-07	0.508 [0.202, 0.663]	0.469 [0.316, 0.765]	0.023 [0.007, 0.056]	10	3	2
LLaMA-3.2-3B	3.28e-06	1.11e-05	1.31e-07	0.227 [0.000, 0.551]	0.764 [0.443, 0.990]	0.009 [0.002, 0.043]	6	3	2

TABLE III. METHOD MEANS AND COMMUNICATION RANGES AT $\alpha=0.1$.

Model	Method	Mean ($\alpha=0.1$)	SD	Comm. range (MB)	Active clients
TinyLlama-1.1B	FedIT	1.6962	0.0038	2320.3–2578.1	9–10
TinyLlama-1.1B	FFA-LoRA	1.7192	0.0026	884.8–983.1	9–10
TinyLlama-1.1B	FLoRA	1.6957	0.0013	2320.3–2578.1	9–10
LLaMA-3.2-3B	FedIT	1.8028	0.0051	2362.5–2625.0	9–10
LLaMA-3.2-3B	FFA-LoRA	1.8355	0.0031	992.2–1102.5	9–10
LLaMA-3.2-3B	FLoRA	1.8049	0.0020	2362.5–2625.0	9–10

Fig. 1. Held-out loss by method: IID (left group) and Dirichlet $\alpha=0.1$ (right group), points colored by partition.

Δ the unpaired design needs fewer draws per method, because the interaction term enters the paired difference twice; where $\hat{\sigma}_{PM}^2$ dominates, pairing on partitions buys little. Pairing is a variance-control device for the shared partition effect, not a substitute for enough independent partitions.

c) Cross-scale reading.: On TinyLlama the partition share is substantial (0.508). On LLaMA-3.2-3B the interaction dominates (0.764) and the partition main-effect CI includes zero, so we do not claim a partition main effect at 3B. Cross-scale CIs on the partition share overlap; the two scales are not distinguishable on that margin. The scales also differ in more than size: TinyLlama-1.1B-Chat-v1.0 is instruction-tuned while LLaMA-3.2-3B is a base model, and base weights are float32 versus float16, so cross-scale differences should not be read as pure scale effects. The rise in LLaMA flip probability from $k=1$ to $k=3$ is a six-partition near-tie artifact, not evidence that more draws hurt. We report the 3B IID floor descriptively only (2 degrees of freedom per method).

d) Limitations.: The study uses two model scales only (1.1B and 3B), one dataset (Dolly), one heterogeneity level ($\alpha=0.1$), three aggregators, 10 clients with full participation, 15 rounds with one local epoch, and held-out next-token loss rather than generation benchmarks. Local updates apply only on complete gradient-accumulation blocks, so clients

holding fewer samples than one block contribute no update; at $\alpha=0.1$ this affected up to two clients per partition, each holding at most 11 samples. LLaMA production holdouts run in float16; the float32 spot-check bound the difference well below the 0.01 effect scale, but the primary 3B numbers remain float16. Only 6 partitions support the 3B grid. The MixedLM partition component did not identify, so moment estimates are primary. Communication is reported within method only. RQ4 is exploratory and unreplicated at 3B. The release covers results and metadata, not adapter weights.

e) No method recommendation.: We make no method recommendation. The separable FFA-LoRA gaps are large and stable; the FedIT–FLoRA gap is not. The contribution is the measurement design and the conditioning of ranking stability on effect size relative to partition-draw noise.

VI. CONCLUSION

At Dirichlet $\alpha=0.1$, the partition draw and its interaction with the method are first-order sources of uncertainty in federated LoRA evaluations. On TinyLlama they account for about half of held-out-loss variance alongside a comparable interaction share; on LLaMA-3.2-3B the interaction dominates and the partition main-effect interval includes zero. Rankings of near-tied methods flip under few draws, while gaps an

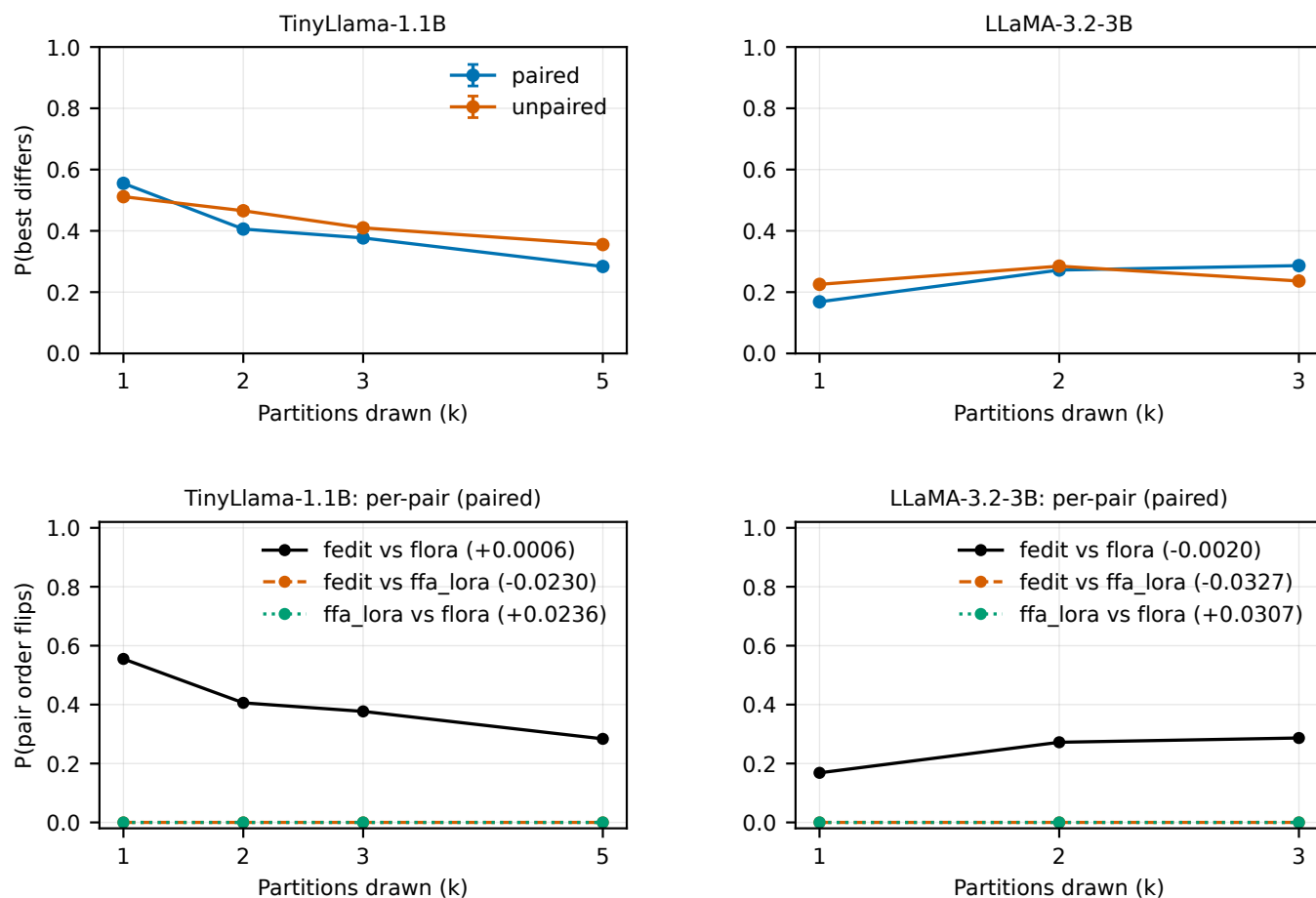


Fig. 2. Ranking-flip probabilities under paired and unpaired protocols.

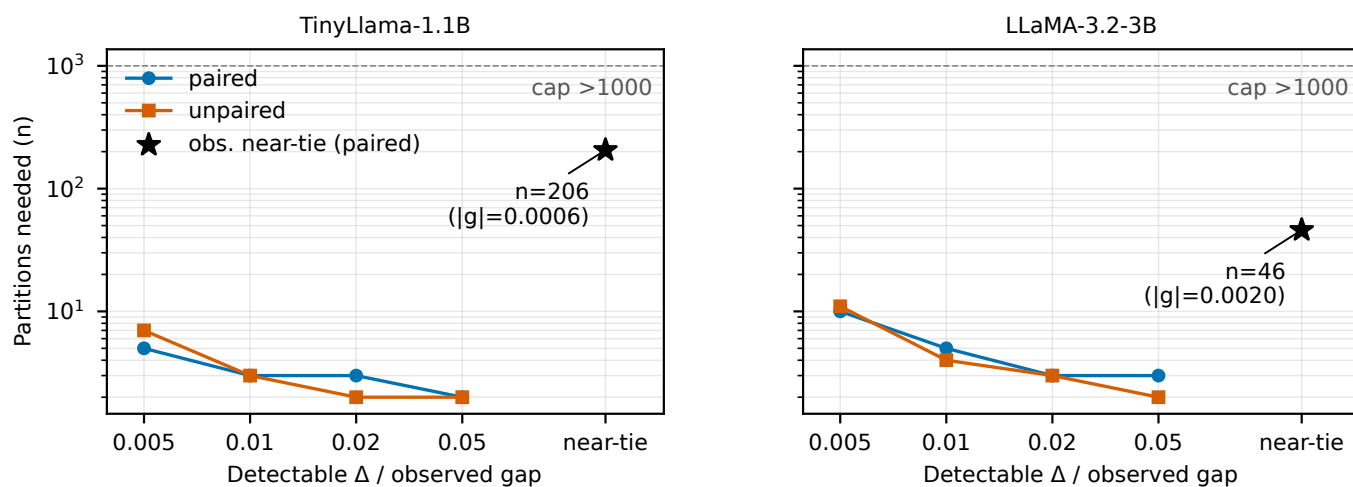


Fig. 3. Partition draws needed for pre-registered deltas and the observed FedIT-FLoRA near-tie (star).

order of magnitude larger than the per-draw paired SD do not. Detecting the observed FedIT-FLoRA near-tie would require hundreds of paired partitions at 1.1B and dozens at 3B. Evaluations should therefore report how many partition

draws were used, pair methods on the same draws, and state the smallest difference the design can detect. We make no method recommendation.

DECLARATIONS

a) *Data and code availability.*: Code and run artifacts will be released upon acceptance.

b) *Funding.*: None.

c) *Conflicts of interest.*: None.

d) *Generative AI.*: The author used Claude (Anthropic) for language editing and drafting support, and AI coding assistants for implementing experiment and analysis scripts. The author designed the study, verified all code and results, and is responsible for all technical content, analysis, and interpretations.

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